A Comparative Study of NLP Methods

Feature-Based · Transformer-Based · Prompt-Based

20 Newsgroups Dataset

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What Is This Project About?

01

TF-IDF & Word2Vec

Simple neural network (MLP) trained with classical text representations

02

BERT Fine-Tuning

Directly fine-tuned bert-base-uncased for 20-class classification

03

MLM Adaptation

Domain-adapted BERT via masked language modeling before classification

04

Llama-3 Prompting

Zero-shot & few-shot evaluation using Meta-Llama-3-8B-Instruct

+ Bonus: QLoRA Parameter-Efficient Fine-Tuning

The Dataset: 20 Newsgroups

9,051
Training Samples

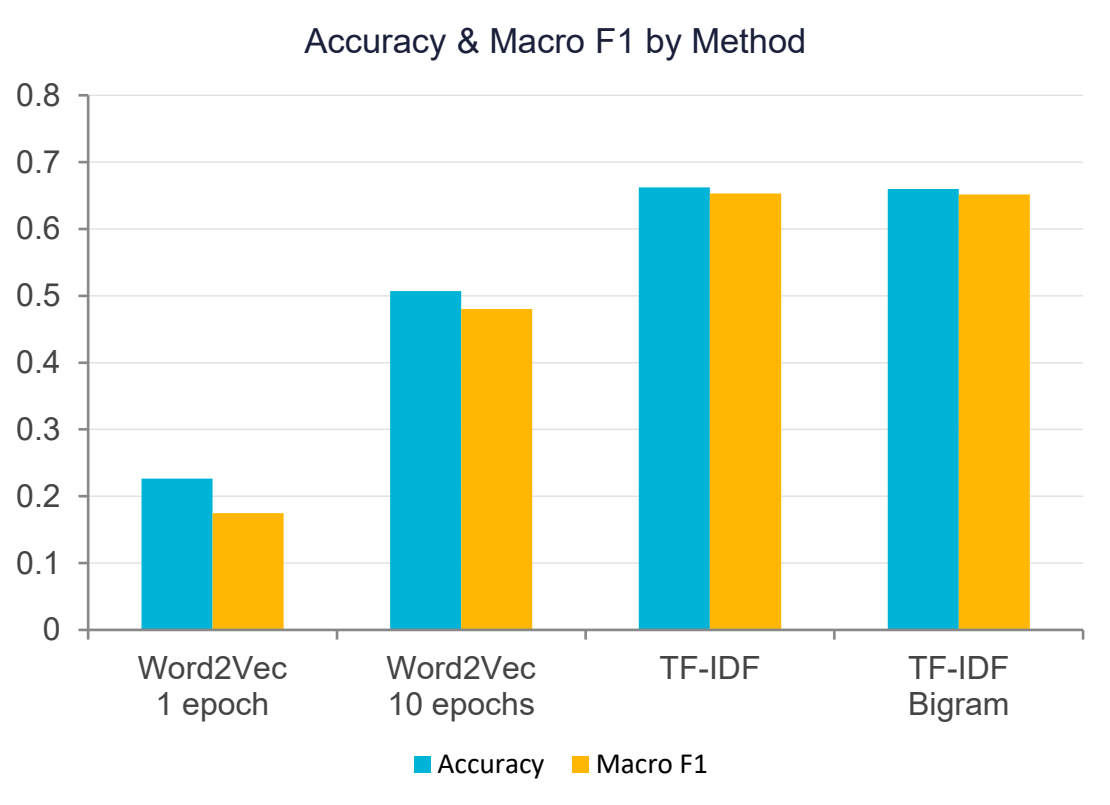
2,263
Validation Samples

7,532
Test Samples

20 Topic Categories

- alt.atheism
- comp.graphics
- comp.os.ms-windows
- comp.sys.ibm.pc
- comp.sys.mac
- comp.windows.x
- misc.forsale
- rec.autos
- rec.motorcycles
- rec.sport.baseball
- rec.sport.hockey
- sci.crypt
- sci.electronics
- sci.med
- sci.space
- soc.religion.christian
- talk.politics.guns
- talk.politics.mideast
- talk.politics.misc
- talk.religion.misc

Part 1 — Classical Representations + MLP



Key Insights

▲ TF-IDF is the clear winner — topic keywords carry strong class signal

↗ Word2Vec jumped from 22% to 51% accuracy with 10× more training

≈ Bigrams added negligible improvement over unigrams

✓ MLP architecture stayed fixed — only the representation changed

Parts 2 & 3 — BERT: Direct vs. MLM-Adapted

Part 2 — Direct Fine-Tuning

- Model: bert-base-uncased
- Max seq length: 64 tokens
- Batch size: 8
- Learning rate: 2×10^{-5}
- Epochs: 2

Accuracy	Macro F1
0.6500	0.6300

MLM boost

Part 3 — MLM then Fine-Tuning

- Base model: bert-base-uncased
- MLM adaptation: 2,000 samples
- MLM epochs: 1
- Then same fine-tuning as Part 2
- Domain-aware BERT

Accuracy	Macro F1
0.6600	0.6390

Part 4 — Llama-3: Zero-Shot vs Few-Shot

Zero-Shot

Model receives only:
- The list of 20 valid labels
- The input text
- Instruction to return one label

No examples provided.

12%
Accuracy

0.1040
Macro F1

Few-Shot

Model receives:
- The list of 20 valid labels
- Labeled example pairs
- The input text

In-context learning — no weight updates.

50%
Accuracy

0.353
Macro F1

Bonus — QLoRA: Parameter-Efficient Fine-Tuning

What is QLoRA?

- LoRA injects trainable low-rank matrices into frozen transformer layers
- QLoRA adds 4-bit quantization so large models fit on consumer GPUs
- Only a fraction of parameters are trained — far cheaper than full fine-tuning
- Ideal for running Llama-3 scale models on limited hardware

Experiment Setup

Model:	Smaller instruct model
Epochs:	1
Environment:	Kaggle GPU
Constraint:	Limited memory

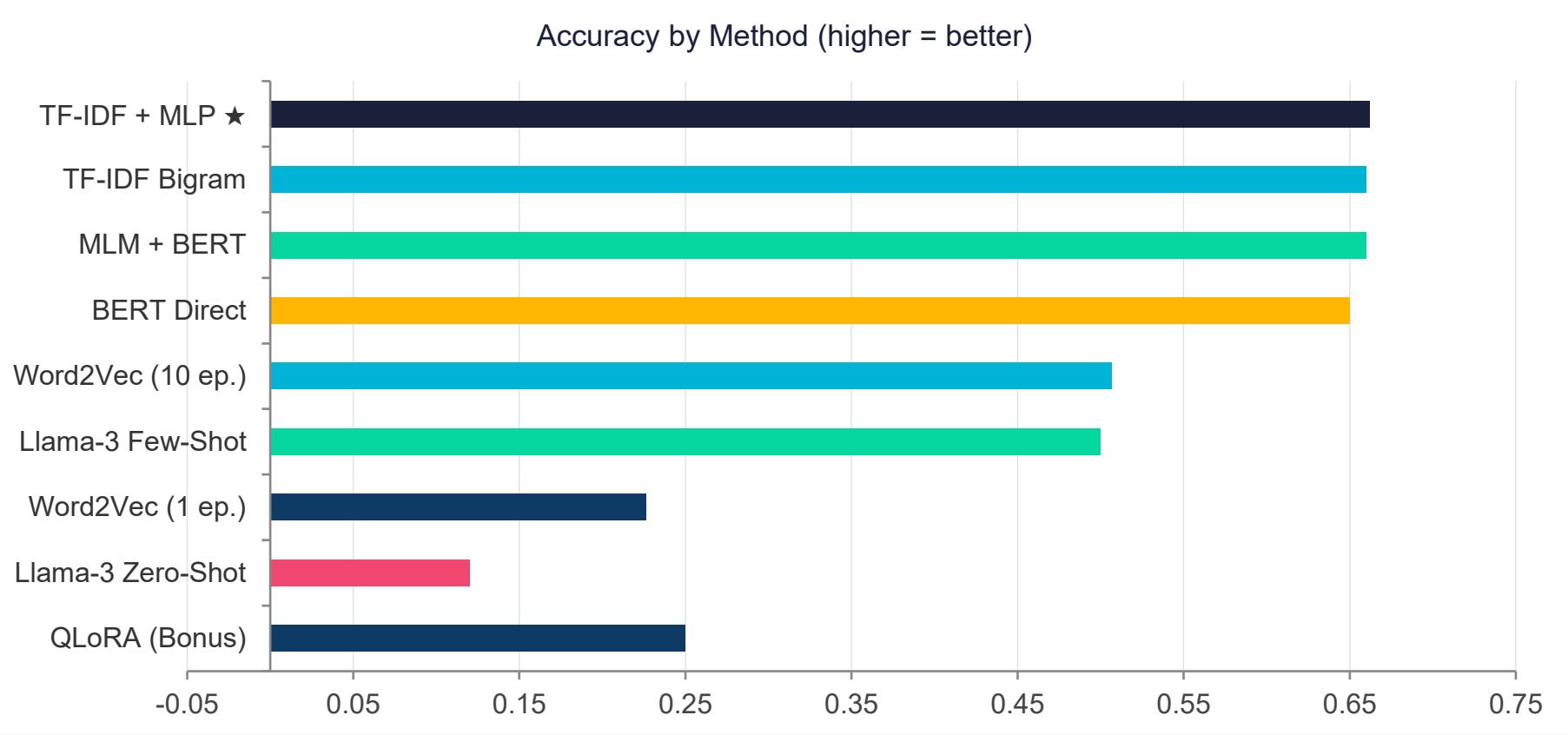
Result

25% Accuracy

0.175 Macro F1

Proof-of-concept only — hardware constraints limited scale, not the method itself.

All Methods — Final Comparison



Key Lessons from the Experiments

01

Right tool for the task

TF-IDF won not because it's the best model in general, but because it perfectly matches a topic-keyword-heavy dataset. Model complexity ≠ guaranteed accuracy.

02

Quality of representation matters

Word2Vec improved from 22% → 51% just by training longer. Embedding quality is as important as the classifier architecture.

03

Domain adaptation helps BERT

MLM pretraining on in-domain text gave BERT a small but consistent boost — even one epoch on 2,000 samples made a difference.

04

Examples unlock LLMs

Llama-3's accuracy jumped from 12% to 50% just by adding in-context examples. Prompt design is as powerful as model selection for inference-only use.

Conclusion

- TF-IDF + MLP achieved the best overall accuracy (66.2%) — simple but powerful for topic tasks
- BERT and MLM-BERT performed competitively, confirming transformer value in NLP
- Few-shot prompting showed the potential of LLMs without any training
- QLoRA demonstrated parameter-efficient fine-tuning is feasible under constraints
- Best method depends on task type, data, and available compute — not just model size